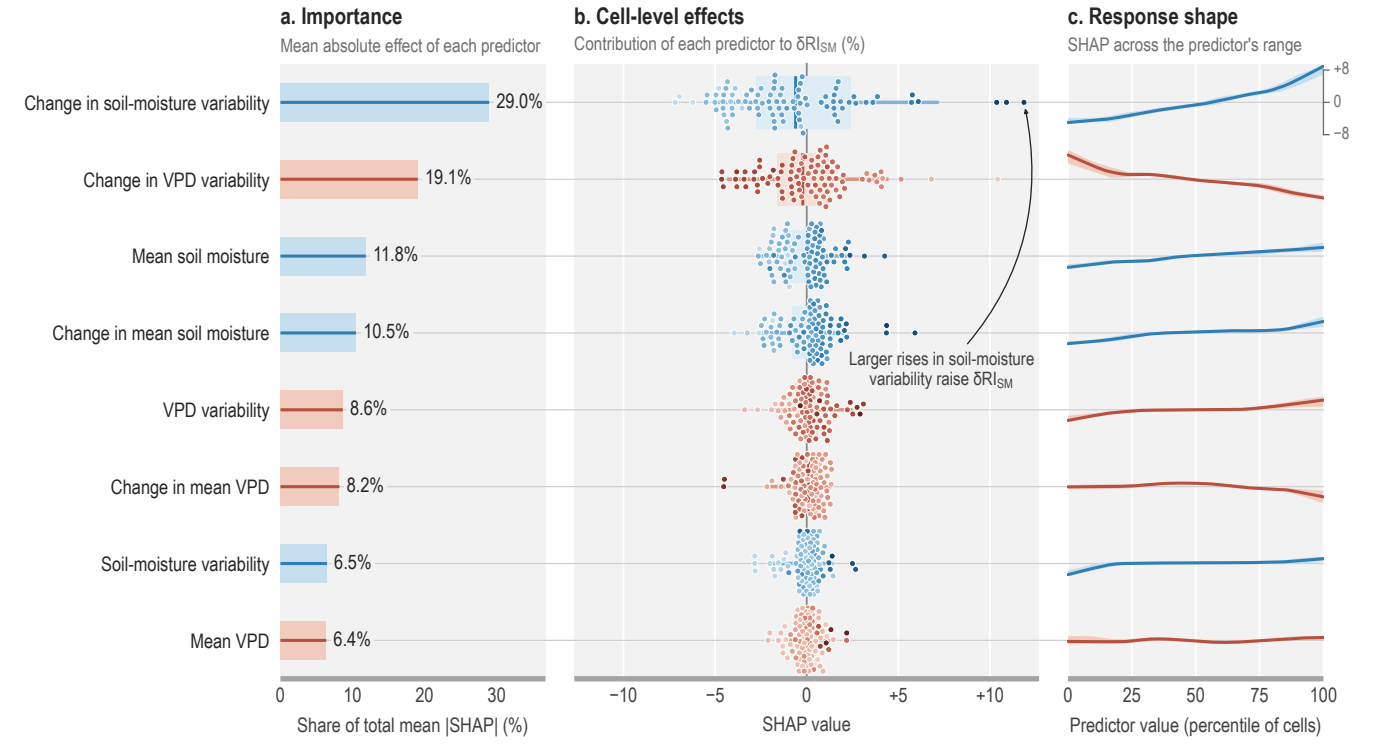
# Soil-moisture importance grows most where soil-moisture variability rises and VPD variability falls.

SHAP attribution of the relative change in soil-moisture importance, $\delta RI_{SM}$. XGBoost, test $R^2$ = 0.43, n = 1,761 cells at 2.5°.




**Predictor groups**

VPD 42.3% | Soil moisture 57.7%

Each segment is one predictor; lengths sum to 100% of mean |SHAP|.

**Cell-level effects**

Bar: median and 25–75% of all cells

Line: 5–95% of all cells

Dots: random 5% of cells (n = 88)

**Dot shade**

Low — High predictor value

**Response shape**

Line: LOWESS fit

Band: local 25–75%

Each row line is SHAP = 0; rows share the scale at the top right.